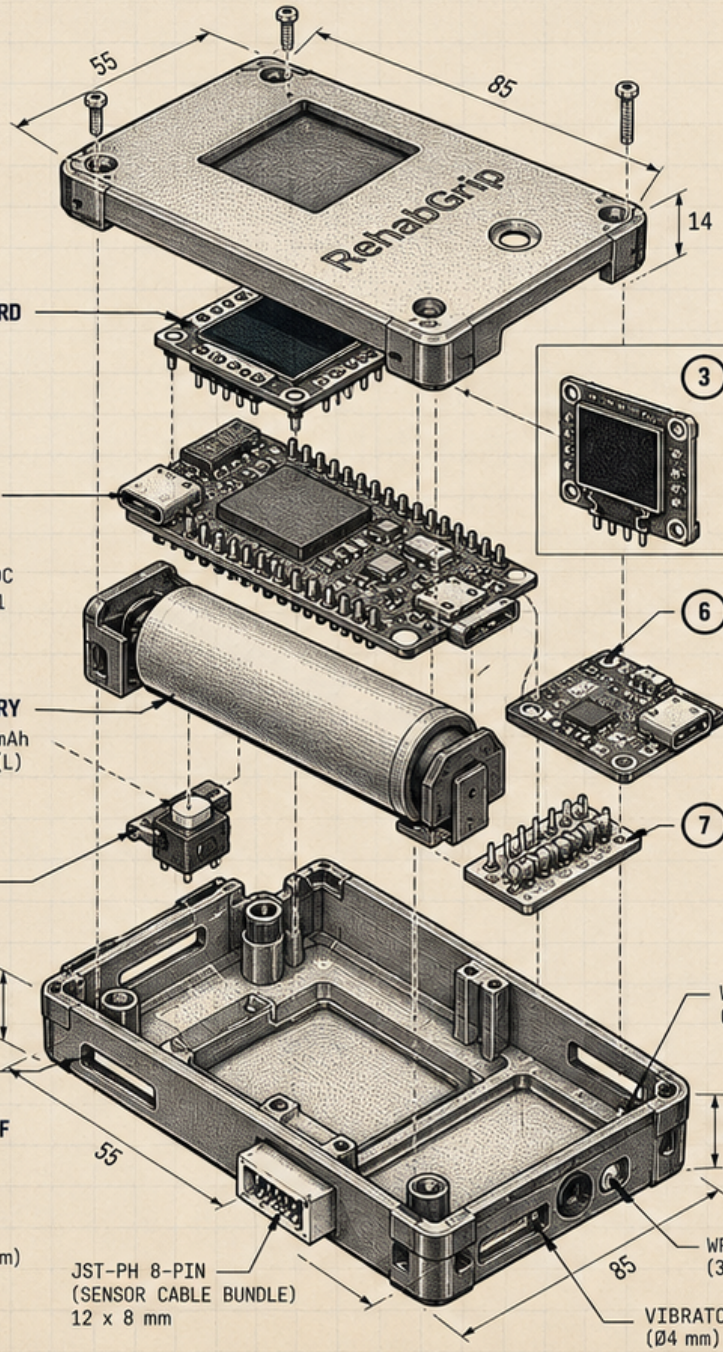
FIGURE 3-2: WRIST MODULE ASSEMBLY — EXPLODED VIEW

RG-2026-ME-003 SHEET 1 OF 1



1 TOP ENCLOSURE HALF
3D PRINTED PLA
85 x 55 x 14 mm
OLED WINDOW (20 x 20 mm)
LED INDICATOR (Ø5 mm)
P/N: RG-ENC-TOP-01

2 OLED DISPLAY BOARD
SH1106 0.96"
27 x 27 x 4 mm
I²C INTERFACE
P/N: RG-DISP-01

3 SH1106 DISPLAY MODULE
132 x 64 PIXEL OLED
I²C BUS (SDA, SCL)
27 x 27 x 4 mm
P/N: RG-OLED-01

4 ESP32-S3 DEVKIT
48 x 26 x 10 mm
USB-C, Wi-Fi / BLE
GPIO32 – 36 FLEX ADC
P/N: RG-ESP32S3-01

5 Li-ion 18650 BATTERY
3.7 V 2600 – 3400 mAh
18 mm (Ø) x 65 mm (L)
P/N: RG-BAT-18650

6 TP4056 CHARGING MODULE
40 x 25 x 10 mm
USB-C (or micro-USB)
5 V input, 3.7 V charge
P/N: RG-CHG-01

7 PERFBOARD (RESISTORS)
FLEX SENSOR VOLTAGE DIVIDERS
10kΩ x 10 (5 fingers + spares)
15 x 10 mm
P/N: RG-PERF-01

8 POWER SWITCH
8 x 8 mm
SPST (ON/OFF)
P/N: RG-SW-01

9 BOTTOM ENCLOSURE HALF
3D PRINTED PLA
85 x 55 x 14 mm
Cable routing channels
Connector cutouts
Wrist strap slots (35 x 4 mm)
P/N: RG-ENC-BOT-01

WRIST STRAP SLOT (35 x 4 mm)
USB-C CUTOUT (FROM TP4056)
JST-PH 8-PIN (SENSOR CABLE BUNDLE) 12 x 8 mm
WRIST STRAP SLOT (35 x 4 mm)
VIBRATOR WIRE GROMMET (Ø4 mm)

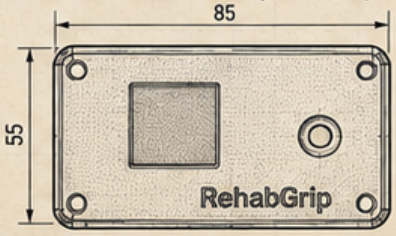
PARTS LIST (BILL OF MATERIALS)

ITEM	QTY	PART NO.	DESCRIPTION
1	1	RG-ENC-TOP-01	Top enclosure, PLA, matte black 85 x 55 x 14 mm, OLED window, LED hole, M2 bosses
2	1	RG-DISP-01	SH1106 OLED display board 0.96", 132 x 64, I²C 27 x 27 x 4 mm
3	1	RG-OLED-01	OLED module (with headers) 132 x 64 pixel, I²C bus
4	1	RG-ESP32S3-01	ESP32-S3 DevKit 48 x 26 x 10 mm, USB-C Wi-Fi / BLE, GPIO32-36
5	1	RG-BAT-18650	18650 Li-ion battery 3.7 V 2600–3400 mAh 18 mm (Ø) x 65 mm (L)
6	1	RG-CHG-01	TP4056 charging module 40 x 25 x 10 mm USB-C (or micro-USB), 5 V input
7	1	RG-PERF-01	Perfboard with 10kΩ resistors FLEX sensor voltage dividers 10kΩ × 10, 15 x 10 mm
8	1	RG-SW-01	Pewer switch, SPST (ON/OFF) 8 x 8 mm
9	1	RG-ENC-BOT-01	Bottom enclosure, PLA 85 x 55 x 14 mm, cable routing, JST-PH port, grommets, strap slots

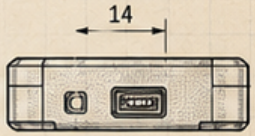
NOTES:

1. All dimensions are in millimeters (mm).
2. Use M2 x 8 mm screws (4 pcs) for enclosure assembly.
3. Ensure proper insulation and strain relief for all cables.
4. JST-PH 8-pin connector used for flex sensors, EMG and auxiliary signals.
5. Enclosure material: 3D printed PLA, matte black, 2.5 mm wall thickness.
6. Refer to Section 4 for wiring details and pin assignments.
7. Battery and charging module must be mounted with insulation and proper clearance.
8. Ensure wrist strap slots are free of burrs for user comfort.

ENCLOSURE DIMENSIONS (TOP VIEW)



SIDE VIEW



REHABGRIP SYSTEMS	WRIST MODULE ASSEMBLY	DWG NO.	REV	SCALE	UNITS	DATE
		RG-2026-ME-003	A	1:1.5	mm	2026